

Confusion Matrix:

[	15624	5	143	72	0	0	0	0	0]
[	147	0	12	2	0	0	0	0	0]
[	1175	0	2398	0	0	0	0	0	0]
[	27	0	20	39	0	0	0	0	0]
[	66	0	22	1	0	0	0	0	0]
[	119	0	10	0	0	0	0	0	0]
[	88	0	0	0	0	0	0	0	0]
[	20	0	0	0	0	0	0	0	0]
[	10	0	0	0	0	0	0	0	0]]

Accuracy : 0.9031

Precision : 0.8824

Recall : 0.9031

F1 Score : 0.8878

Classification Report:

	precision	recall	f1-score	support
0	0.90	0.99	0.94	15844
1	0.00	0.00	0.00	161
2	0.92	0.67	0.78	3573
3	0.34	0.45	0.39	86
4	0.00	0.00	0.00	89
5	0.00	0.00	0.00	129
6	0.00	0.00	0.00	88
7	0.00	0.00	0.00	20
8	0.00	0.00	0.00	10
accuracy			0.90	20000
macro avg	0.24	0.23	0.23	20000
weighted avg	0.88	0.90	0.89	20000

Confusion Matrix:

[[3157	1	2	1	0	0	0	3	1]
[2	26	1	0	0	0	0	0	0]
[9	2	711	0	0	0	0	0	0]
[1	0	0	15	0	0	0	0	0]
[0	0	0	0	19	0	0	0	0]
[0	0	0	0	0	27	0	0	0]
[1	0	0	0	0	0	16	0	0]
[3	0	0	0	0	0	0	0	0]
[1	0	0	0	0	0	0	1	0]]

Accuracy : 0.9928

Precision : 0.9928

Recall : 0.9928

F1 Score : 0.9927

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	1.00	3165
1	0.90	0.90	0.90	29
2	1.00	0.98	0.99	722
3	0.94	0.94	0.94	16
4	1.00	1.00	1.00	19
5	1.00	1.00	1.00	27
6	1.00	0.94	0.97	17
7	0.00	0.00	0.00	3
8	0.00	0.00	0.00	2
accuracy			0.99	4000
macro avg	0.76	0.75	0.75	4000
weighted avg	0.99	0.99	0.99	4000

Confusion Matrix:

[	[	3158	1	5	1	0	0	0	0	0]
[	2	27	0	0	0	0	0	0	0	0]
[	9	1	712	0	0	0	0	0	0	0]
[	1	0	0	15	0	0	0	0	0	0]
[	0	0	0	0	19	0	0	0	0	0]
[	0	0	0	0	0	27	0	0	0	0]
[	0	0	0	0	0	0	17	0	0	0]
[	3	0	0	0	0	0	0	0	0	0]
[	2	0	0	0	0	0	0	0	0	0]]

Accuracy : 0.9938

Precision : 0.9925

Recall : 0.9938

F1 Score : 0.9931

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	1.00	3165
1	0.93	0.93	0.93	29
2	0.99	0.99	0.99	722
3	0.94	0.94	0.94	16
4	1.00	1.00	1.00	19
5	1.00	1.00	1.00	27
6	1.00	1.00	1.00	17
7	0.00	0.00	0.00	3
8	0.00	0.00	0.00	2
accuracy			0.99	4000
macro avg	0.76	0.76	0.76	4000
weighted avg	0.99	0.99	0.99	4000

Confusion Matrix:

[	2994	0	165	6	0	0	0	0	0]
[	28	0	1	0	0	0	0	0	0]
[	81	0	641	0	0	0	0	0	0]
[	11	0	5	0	0	0	0	0	0]
[	10	0	9	0	0	0	0	0	0]
[	27	0	0	0	0	0	0	0	0]
[	17	0	0	0	0	0	0	0	0]
[	3	0	0	0	0	0	0	0	0]
[	2	0	0	0	0	0	0	0	0]]

Accuracy : 0.9087

Precision : 0.8875

Recall : 0.9087

F1 Score : 0.8975

Classification Report:

	precision	recall	f1-score	support
0	0.94	0.95	0.94	3165
1	0.00	0.00	0.00	29
2	0.78	0.89	0.83	722
3	0.00	0.00	0.00	16
4	0.00	0.00	0.00	19
5	0.00	0.00	0.00	27
6	0.00	0.00	0.00	17
7	0.00	0.00	0.00	3
8	0.00	0.00	0.00	2
accuracy			0.91	4000
macro avg	0.19	0.20	0.20	4000
weighted avg	0.89	0.91	0.90	4000

0.9877
Confusion Matrix:

```
[[3140    3    9    3    2    1    0    1    6]
 [    3   25    1    0    0    0    0    0    0]
 [   13    1   70    0    1    0    0    0    0]
 [    1    0    0   15    0    0    0    0    0]
 [    0    0    0    0   19    0    0    0    0]
 [    1    0    0    0    0   26    0    0    0]
 [    1    0    0    0    0    0   16    0    0]
 [    3    0    0    0    0    0    0    0    0]
 [    0    0    0    0    0    0    0    0    2]]
```

Accuracy : 0.9875

Precision : 0.9883

Recall : 0.9875

F1 Score : 0.9877

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	3165
1	0.86	0.86	0.86	29
2	0.99	0.98	0.98	722
3	0.83	0.94	0.88	16
4	0.86	1.00	0.93	19
5	0.96	0.96	0.96	27
6	1.00	0.94	0.97	17
7	0.00	0.00	0.00	3
8	0.25	1.00	0.40	2
accuracy			0.99	4000
macro avg	0.75	0.85	0.78	4000
weighted avg	0.99	0.99	0.99	4000

```
[logging] [warning] stopped training because there are no
Confusion Matrix:

[[3007    9  107    1    8   33    0    0    0]
 [    4   20    3    0    2    0    0    0    0]
 [  161    5  545    0    0   11    0    0    0]
 [   10    0    6    0    0    0    0    0    0]
 [    5    0    5    0    9    0    0    0    0]
 [   27    0    0    0    0    0    0    0    0]
 [   17    0    0    0    0    0    0    0    0]
 [    3    0    0    0    0    0    0    0    0]
 [    2    0    0    0    0    0    0    0    0]]
```

Accuracy : 0.8952

Precision : 0.8895

Recall : 0.8952

F1 Score : 0.8920

Classification Report:

	precision	recall	f1-score	support
0	0.93	0.95	0.94	3165
1	0.59	0.69	0.63	29
2	0.82	0.75	0.79	722
3	0.00	0.00	0.00	16
4	0.47	0.47	0.47	19
5	0.00	0.00	0.00	27
6	0.00	0.00	0.00	17
7	0.00	0.00	0.00	3
8	0.00	0.00	0.00	2
accuracy			0.90	4000
macro avg	0.31	0.32	0.31	4000
weighted avg	0.89	0.90	0.89	4000

```
In [21]: -|
```

Confusion Matrix:

[	3154	0	4	0	0	0	0	5	2]
[	0	29	0	0	0	0	0	0	0]
[	3	1	718	0	0	0	0	0	0]
[	1	0	0	15	0	0	0	0	0]
[	0	0	0	0	19	0	0	0	0]
[	0	0	0	0	0	27	0	0	0]
[	0	0	0	0	0	0	17	0	0]
[	3	0	0	0	0	0	0	0	0]
[	0	0	0	0	0	0	0	2	0]]

Accuracy : 0.9948

Precision : 0.9958

Recall : 0.9948

F1 Score : 0.9952

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	3165
1	0.97	1.00	0.98	29
2	0.99	0.99	0.99	722
3	1.00	0.94	0.97	16
4	1.00	1.00	1.00	19
5	1.00	1.00	1.00	27
6	1.00	1.00	1.00	17
7	0.00	0.00	0.00	3
8	0.00	0.00	0.00	2
accuracy			0.99	4000
macro avg	0.77	0.77	0.77	4000
weighted avg	1.00	0.99	1.00	4000

Confusion Matrix:

```
[[3157    0    3    1    0    0    0    3    1]
 [    0   29    0    0    0    0    0    0    0]
 [    7    1  714    0    0    0    0    0    0]
 [    1    0    0   15    0    0    0    0    0]
 [    0    0    0    0   19    0    0    0    0]
 [    0    0    0    0    0   27    0    0    0]
 [    0    0    0    0    0    0   17    0    0]
 [    3    0    0    0    0    0    0    0    0]
 [    1    0    0    0    0    0    0    1    0]]
```

Accuracy : 0.9945

Precision : 0.9945

Recall : 0.9945

F1 Score : 0.9945

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	3165
1	0.97	1.00	0.98	29
2	1.00	0.99	0.99	722
3	0.94	0.94	0.94	16
4	1.00	1.00	1.00	19
5	1.00	1.00	1.00	27
6	1.00	1.00	1.00	17
7	0.00	0.00	0.00	3
8	0.00	0.00	0.00	2
accuracy			0.99	4000
macro avg	0.77	0.77	0.77	4000
weighted avg	0.99	0.99	0.99	4000

```
In [3]: %runfile 'D:/python project/sgbc.py' --wdir
```

Confusion Matrix:

```
[[15839      3     14      3      0      0      0      5      0      0]
 [      6    144      5      0      0      0      0      0      0      0]
 [      5      0   3560      0      0      0      0      0      0      0]
 [      2      0      0     82      0      0      0      0      0      0]
 [      1      0      0      2    86      0      0      0      0      0]
 [      0      0      0      0      0    119      0      0      0      0]
 [      2      0      0      0      0      0     90      0      0      0]
 [      4      0      0      0      0      0      0     16      0      2]
 [      1      0      0      0      0      0      0      0      0      0]
 [      1      0      0      0      0      0      0      7      0      1]]
```

Accuracy : 0.9969

Precision : 0.9967

Recall : 0.9969

F1 Score : 0.9967

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	15864
1	0.98	0.93	0.95	155
2	0.99	1.00	1.00	3565
3	0.94	0.98	0.96	84
4	1.00	0.97	0.98	89
5	1.00	1.00	1.00	119
6	1.00	0.98	0.99	92
7	0.57	0.73	0.64	22
8	0.00	0.00	0.00	1
9	0.33	0.11	0.17	9
accuracy			1.00	20000
macro avg	0.78	0.77	0.77	20000
weighted avg	1.00	1.00	1.00	20000

Confusion Matrix:

```
[[3023    1  141    0    0    0    0    0    0]
 [   26    0    3    0    0    0    0    0    0]
 [  158    0  564    0    0    0    0    0    0]
 [   11    0    5    0    0    0    0    0    0]
 [   19    0    0    0    0    0    0    0    0]
 [   27    0    0    0    0    0    0    0    0]
 [   17    0    0    0    0    0    0    0    0]
 [    3    0    0    0    0    0    0    0    0]
 [    2    0    0    0    0    0    0    0    0]]
```

Accuracy : 0.8968

Precision : 0.8707

Recall : 0.8968

F1 Score : 0.8835

Classification Report:

	precision	recall	f1-score	support
0	0.92	0.96	0.94	3165
1	0.00	0.00	0.00	29
2	0.79	0.78	0.79	722
3	0.00	0.00	0.00	16
4	0.00	0.00	0.00	19
5	0.00	0.00	0.00	27
6	0.00	0.00	0.00	17
7	0.00	0.00	0.00	3
8	0.00	0.00	0.00	2
accuracy			0.90	4000
macro avg	0.19	0.19	0.19	4000
weighted avg	0.87	0.90	0.88	4000

```
In [1]: %runfile 'D:/python project/ada boost classifier.py' --wdir
Confusion Matrix:
```

```
[[15652      3    209      0      0      0      0      0      0      0]
 [   101      6     48      0      0      0      0      0      0      0]
 [   506      2   3057      0      0      0      0      0      0      0]
 [    84      0      0      0      0      0      0      0      0      0]
 [    75      0      0      0    14      0      0      0      0      0]
 [   119      0      0      0      0      0      0      0      0      0]
 [    92      0      0      0      0      0      0      0      0      0]
 [    22      0      0      0      0      0      0      0      0      0]
 [     1      0      0      0      0      0      0      0      0      0]
 [     9      0      0      0      0      0      0      0      0      0]]
```

Accuracy : 0.9365

Precision : 0.9183

Recall : 0.9365

F1 Score : 0.9236

Classification Report:

	precision	recall	f1-score	support
0	0.94	0.99	0.96	15864
1	0.55	0.04	0.07	155
2	0.92	0.86	0.89	3565
3	0.00	0.00	0.00	84
4	1.00	0.16	0.27	89
5	0.00	0.00	0.00	119
6	0.00	0.00	0.00	92
7	0.00	0.00	0.00	22
8	0.00	0.00	0.00	1
9	0.00	0.00	0.00	9
accuracy			0.94	20000
macro avg	0.34	0.20	0.22	20000
weighted avg	0.92	0.94	0.92	20000

```
[...8] [...8] [...8] [...8] [...8] [...8] [...8] [...8] [...8] [...8]
```

Confusion Matrix:

```
[[15437    5   228   21   24    3   122   23    1    0]
 [   68   20    25    0    0   14    0   28    0    0]
 [  266    1  3180   20    7   11   20   55    5    0]
 [   65    0   17    2    0    0    0    0    0    0]
 [   45    0   30    0   14    0    0    0    0    0]
 [  104    0    0    0    0    6    9    0    0    0]
 [   92    0    0    0    0    0    0    0    0    0]
 [   15    0    0    0    7    0    0    0    0    0]
 [    1    0    0    0    0    0    0    0    0    0]
 [    8    0    0    0    1    0    0    0    0    0]]
```

Accuracy : 0.9329

Precision : 0.9318

Recall : 0.9329

F1 Score : 0.9302

Classification Report:

	precision	recall	f1-score	support
0	0.96	0.97	0.97	15864
1	0.77	0.13	0.22	155
2	0.91	0.89	0.90	3565
3	0.05	0.02	0.03	84
4	0.26	0.16	0.20	89
5	0.18	0.05	0.08	119
6	0.00	0.00	0.00	92
7	0.00	0.00	0.00	22
8	0.00	0.00	0.00	1
9	0.00	0.00	0.00	9
accuracy			0.93	20000
macro avg	0.31	0.22	0.24	20000
weighted avg	0.93	0.93	0.93	20000

```
In [1]: %runfile 'D:/python project/ada boost classifier.py' --wdir
Confusion Matrix:
```

```
[[15652      3    209      0      0      0      0      0      0      0]
 [   101      6     48      0      0      0      0      0      0      0]
 [   506      2   3057      0      0      0      0      0      0      0]
 [    84      0      0      0      0      0      0      0      0      0]
 [    75      0      0      0    14      0      0      0      0      0]
 [   119      0      0      0      0      0      0      0      0      0]
 [    92      0      0      0      0      0      0      0      0      0]
 [    22      0      0      0      0      0      0      0      0      0]
 [     1      0      0      0      0      0      0      0      0      0]
 [     9      0      0      0      0      0      0      0      0      0]]
```

Accuracy : 0.9365

Precision : 0.9183

Recall : 0.9365

F1 Score : 0.9236

Classification Report:

	precision	recall	f1-score	support
0	0.94	0.99	0.96	15864
1	0.55	0.04	0.07	155
2	0.92	0.86	0.89	3565
3	0.00	0.00	0.00	84
4	1.00	0.16	0.27	89
5	0.00	0.00	0.00	119
6	0.00	0.00	0.00	92
7	0.00	0.00	0.00	22
8	0.00	0.00	0.00	1
9	0.00	0.00	0.00	9
accuracy			0.94	20000
macro avg	0.34	0.20	0.22	20000
weighted avg	0.92	0.94	0.92	20000

Confusion Matrix:

```
[[15843      1     15      1      0      1      1      2      0      0]
 [      6    147      2      0      0      0      0      0      0      0]
 [     10      0   3555      0      0      0      0      0      0      0]
 [      1      0      0     83      0      0      0      0      0      0]
 [      0      0      0      2    87      0      0      0      0      0]
 [      2      0      0      0      0    117      0      0      0      0]
 [      2      0      0      0      0      0     90      0      0      0]
 [      8      0      0      0      0      0      0     12      0      2]
 [      1      0      0      0      0      0      0      0      0      0]
 [      3      0      0      0      0      0      0      1      5      0]]
```

Accuracy : 0.9967

Precision : 0.9962

Recall : 0.9967

F1 Score : 0.9965

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	15864
1	0.99	0.95	0.97	155
2	1.00	1.00	1.00	3565
3	0.97	0.99	0.98	84
4	1.00	0.98	0.99	89
5	0.99	0.98	0.99	119
6	0.98	0.98	0.98	92
7	0.63	0.55	0.59	22
8	0.00	0.00	0.00	1
9	0.00	0.00	0.00	9
accuracy			1.00	20000
macro avg	0.76	0.74	0.75	20000
weighted avg	1.00	1.00	1.00	20000

Confusion Matrix:

```
[[ 94  0  0  0  0  0  0 562 2509  0]
 [  1  0  0  0  0  0  0  6  22  0]
 [  0  0  0  0  0  0  0  9  713  0]
 [  2  0  0  0  0  0  0 13   1  0]
 [  0  0  0  0  0  0  0 14   5  0]
 [  0  0  0  0  0  0  0 15  12  0]
 [  1  0  0  0  0  0  0 10   6  0]
 [  0  0  0  0  0  0  0  1   2  0]
 [  0  0  0  0  0  0  0  0   0  2]]
```

Accuracy : 0.0270

Precision : 0.7595

Recall : 0.0270

F1 Score : 0.0462

Classification Report:

	precision	recall	f1-score	support
0	0.96	0.03	0.06	3165
1	0.00	0.00	0.00	29
2	0.00	0.00	0.00	722
3	0.00	0.00	0.00	16
4	0.00	0.00	0.00	19
5	0.00	0.00	0.00	27
6	0.02	0.59	0.03	17
7	0.00	0.67	0.00	3
8	1.00	1.00	1.00	2
accuracy			0.03	4000
macro avg	0.22	0.25	0.12	4000
weighted avg	0.76	0.03	0.05	4000

```
In [3]: %runfile 'D:/python project/sgbc.py' --wdir
```

Confusion Matrix:

```
[[15839      3     14      3      0      0      0      5      0      0]
 [      6    144      5      0      0      0      0      0      0      0]
 [      5      0   3560      0      0      0      0      0      0      0]
 [      2      0      0     82      0      0      0      0      0      0]
 [      1      0      0      2    86      0      0      0      0      0]
 [      0      0      0      0      0    119      0      0      0      0]
 [      2      0      0      0      0      0     90      0      0      0]
 [      4      0      0      0      0      0      0     16      0      2]
 [      1      0      0      0      0      0      0      0      0      0]
 [      1      0      0      0      0      0      0      7      0      1]]
```

Accuracy : 0.9969

Precision : 0.9967

Recall : 0.9969

F1 Score : 0.9967

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	15864
1	0.98	0.93	0.95	155
2	0.99	1.00	1.00	3565
3	0.94	0.98	0.96	84
4	1.00	0.97	0.98	89
5	1.00	1.00	1.00	119
6	1.00	0.98	0.99	92
7	0.57	0.73	0.64	22
8	0.00	0.00	0.00	1
9	0.33	0.11	0.17	9
accuracy			1.00	20000
macro avg	0.78	0.77	0.77	20000
weighted avg	1.00	1.00	1.00	20000

Confusion Matrix:

```
[[15843      1     15      1      0      1      1      2      0      0]
 [      6    147      2      0      0      0      0      0      0      0]
 [     10      0   3555      0      0      0      0      0      0      0]
 [      1      0      0     83      0      0      0      0      0      0]
 [      0      0      0      2     87      0      0      0      0      0]
 [      2      0      0      0      0    117      0      0      0      0]
 [      2      0      0      0      0      0     90      0      0      0]
 [      8      0      0      0      0      0      0     12      0      2]
 [      1      0      0      0      0      0      0      0      0      0]
 [      3      0      0      0      0      0      0      1      5      0]]
```

Accuracy : 0.9967

Precision : 0.9962

Recall : 0.9967

F1 Score : 0.9965

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	15864
1	0.99	0.95	0.97	155
2	1.00	1.00	1.00	3565
3	0.97	0.99	0.98	84
4	1.00	0.98	0.99	89
5	0.99	0.98	0.99	119
6	0.98	0.98	0.98	92
7	0.63	0.55	0.59	22
8	0.00	0.00	0.00	1
9	0.00	0.00	0.00	9
accuracy			1.00	20000
macro avg	0.76	0.74	0.75	20000
weighted avg	1.00	1.00	1.00	20000

```
[1 8 0 0 0 0 0 0 0 0] [1 0 0 0 0 0 0 0 0 0] [1 0 0 0 0 0 0 0 0 0] [1 0 0 0 0 0 0 0 0 0] [1 0 0 0 0 0 0 0 0 0]
```

Confusion Matrix:

```
[[15437    5   228   21   24    3   122   23    1    0]
 [   68   20    25    0    0   14    0   28    0    0]
 [  266    1  3180   20    7   11   20   55    5    0]
 [   65    0   17    2    0    0    0    0    0    0]
 [   45    0   30    0   14    0    0    0    0    0]
 [  104    0    0    0    0    6    9    0    0    0]
 [   92    0    0    0    0    0    0    0    0    0]
 [   15    0    0    0    7    0    0    0    0    0]
 [    1    0    0    0    0    0    0    0    0    0]
 [    8    0    0    0    1    0    0    0    0    0]]
```

Accuracy : 0.9329

Precision : 0.9318

Recall : 0.9329

F1 Score : 0.9302

Classification Report:

	precision	recall	f1-score	support
0	0.96	0.97	0.97	15864
1	0.77	0.13	0.22	155
2	0.91	0.89	0.90	3565
3	0.05	0.02	0.03	84
4	0.26	0.16	0.20	89
5	0.18	0.05	0.08	119
6	0.00	0.00	0.00	92
7	0.00	0.00	0.00	22
8	0.00	0.00	0.00	1
9	0.00	0.00	0.00	9
accuracy			0.93	20000
macro avg	0.31	0.22	0.24	20000
weighted avg	0.93	0.93	0.93	20000

```
In [1]: %runfile 'D:/python project/ada boost classifier.py' --wdir
Confusion Matrix:
```

```
[[15652      3    209      0      0      0      0      0      0      0]
 [   101      6     48      0      0      0      0      0      0      0]
 [   506      2   3057      0      0      0      0      0      0      0]
 [    84      0      0      0      0      0      0      0      0      0]
 [    75      0      0      0     14      0      0      0      0      0]
 [   119      0      0      0      0      0      0      0      0      0]
 [    92      0      0      0      0      0      0      0      0      0]
 [    22      0      0      0      0      0      0      0      0      0]
 [     1      0      0      0      0      0      0      0      0      0]
 [     9      0      0      0      0      0      0      0      0      0]]
```

Accuracy : 0.9365

Precision : 0.9183

Recall : 0.9365

F1 Score : 0.9236

Classification Report:

	precision	recall	f1-score	support
0	0.94	0.99	0.96	15864
1	0.55	0.04	0.07	155
2	0.92	0.86	0.89	3565
3	0.00	0.00	0.00	84
4	1.00	0.16	0.27	89
5	0.00	0.00	0.00	119
6	0.00	0.00	0.00	92
7	0.00	0.00	0.00	22
8	0.00	0.00	0.00	1
9	0.00	0.00	0.00	9
accuracy			0.94	20000
macro avg	0.34	0.20	0.22	20000
weighted avg	0.92	0.94	0.92	20000

=====

OUTPUT #2: 9. XGBoost

Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

[[229959	89	1449	71	71	57	1	0	27]
[220	1850	17	2	0	0	0	0	1]
[1274	0	45372	2	6	0	0	0	0]
[108	0	1	981	16	0	0	0	0]
[235	1	2	13	897	0	0	0	0]
[69	0	0	0	0	1512	0	0	1]
[1	0	0	0	0	0	1	0	0]
[3	0	0	0	0	0	0	2	0]
[445	0	2	0	0	0	0	0	685]]

Accuracy : 0.9853

Precision : 0.9852

Recall : 0.9853

F1 Score : 0.9851

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	231724
1	0.95	0.89	0.92	2090
2	0.97	0.97	0.97	46654
3	0.92	0.89	0.90	1106
4	0.91	0.78	0.84	1148
5	0.96	0.96	0.96	1582
6	0.50	0.50	0.50	2
7	1.00	0.40	0.57	5
8	0.96	0.61	0.74	1132
accuracy			0.99	285443

OUTPUT #4: 11. CatBoost

Parameters: test_size = 0.2 | PCA n_components = 5

Confusion Matrix:

```
[[228923    140    2159    108    163    135         0         0     96]
 [   402   1623      54      2      0      8         0         0      1]
 [  2398        0  44224        0    27      1         0         0      4]
 [   317        0        0   773    16      0         0         0      0]
 [   316        1     20      8   803      0         0         0      0]
 [   183        0      3      0      0  1396         0         0      0]
 [      2        0      0      0      0      0         0         0      0]
 [      5        0      0      0      0      0         0         0      0]
 [   494        0      3      0      0      1         0         0   634]]
```

Accuracy : 0.9752

Precision : 0.9748

Recall : 0.9752

F1 Score : 0.9748

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.99	0.99	231724
1	0.92	0.78	0.84	2090
2	0.95	0.95	0.95	46654
3	0.87	0.70	0.77	1106
4	0.80	0.70	0.74	1148
5	0.91	0.88	0.89	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.86	0.56	0.68	1132
accuracy			0.98	285443
macro avg	0.70	0.62	0.65	285443

OUTPUT #4: 11. CatBoost

Parameters: test_size = 0.2 | PCA n_components = 5

Confusion Matrix:

```
[[228923    140    2159    108    163    135         0         0     96]
 [   402   1623      54      2      0      8         0         0      1]
 [  2398        0  44224        0    27      1         0         0      4]
 [   317        0        0   773    16      0         0         0      0]
 [   316        1     20      8   803      0         0         0      0]
 [   183        0      3      0      0  1396         0         0      0]
 [      2        0      0      0      0      0         0         0      0]
 [      5        0      0      0      0      0         0         0      0]
 [   494        0      3      0      0      1         0         0   634]]
```

Accuracy : 0.9752

Precision : 0.9748

Recall : 0.9752

F1 Score : 0.9748

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.99	0.99	231724
1	0.92	0.78	0.84	2090
2	0.95	0.95	0.95	46654
3	0.87	0.70	0.77	1106
4	0.80	0.70	0.74	1148
5	0.91	0.88	0.89	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.86	0.56	0.68	1132
accuracy			0.98	285443
macro avg	0.70	0.62	0.65	285443

OUTPUT #3: 10. LightGBM

Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

```
[[221702    344    7998    591    548    90      0    21    430]
 [   362   1689     20      0    10      0      0      0      9]
 [   3525     64   42003    953    50      0      0    59      0]
 [    359    136       0    595    14      0      0      2      0]
 [   1103      4     12     10    18      0      0      0      1]
 [    207      0      2      0    10   1363      0      0      0]
 [      1      0      0      1      0      0      0      0      0]
 [      5      0      0      0      0      0      0      0      0]
 [    518     20      3     20    490     33      0      0    48]]
```

Accuracy : 0.9369

Precision : 0.9394

Recall : 0.9369

F1 Score : 0.9377

Classification Report:

	precision	recall	f1-score	support
0	0.97	0.96	0.96	231724
1	0.75	0.81	0.78	2090
2	0.84	0.90	0.87	46654
3	0.27	0.54	0.36	1106
4	0.02	0.02	0.02	1148
5	0.92	0.86	0.89	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.10	0.04	0.06	1132
accuracy			0.94	285443
macro avg	0.43	0.46	0.44	285443

=====

OUTPUT #5: 12. AdaBoost

Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

```
[[228792      0    2932      0      0      0      0      0      0]
 [   1923      0     167      0      0      0      0      0      0]
 [ 19195      0   27459      0      0      0      0      0      0]
 [    916      0     190      0      0      0      0      0      0]
 [    926      0     222      0      0      0      0      0      0]
 [   1582      0        0      0      0      0      0      0      0]
 [        2      0        0      0      0      0      0      0      0]
 [        4      0        1      0      0      0      0      0      0]
 [   1130      0        2      0      0      0      0      0      0]]
```

Accuracy : 0.8977

Precision : 0.8748

Recall : 0.8977

F1 Score : 0.8797

Classification Report:

	precision	recall	f1-score	support
0	0.90	0.99	0.94	231724
1	0.00	0.00	0.00	2090
2	0.89	0.59	0.71	46654
3	0.00	0.00	0.00	1106
4	0.00	0.00	0.00	1148
5	0.00	0.00	0.00	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.00	0.00	0.00	1132
accuracy			0.90	285443
macro avg	0.20	0.18	0.18	285443

OUTPUT #6: 8. Gradient Boosting

Parameters: test_size = 0.4 | PCA n_components = 10

Confusion Matrix:

```
[[458655    275    3493    293    104    103         0    571    172]
 [   180   3958         6         3         0         0         0         0         0]
 [   4162         3   88643         8         4         0         0        86         0]
 [   156         3        12   1982        63         0         0         0         9]
 [    449         1         1        13   1876         1         0         0         0]
 [     84         0         2         0         0   3072         0         0         2]
 [      4         0         0         0         0         0         0         0         0]
 [     14         0         0         0         0         0         0         0         0]
 [    399         0         2         0         0         1         0         0    2021]]
```

Accuracy : 0.9813

Precision : 0.9823

Recall : 0.9813

F1 Score : 0.9818

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	463666
1	0.93	0.95	0.94	4147
2	0.96	0.95	0.96	92906
3	0.86	0.89	0.88	2225
4	0.92	0.80	0.86	2341
5	0.97	0.97	0.97	3160
6	0.00	0.00	0.00	4
7	0.00	0.00	0.00	14
8	0.92	0.83	0.87	2423
accuracy			0.98	570886
macro avg	0.73	0.71	0.72	570886

=====

OUTPUT #5: 12. AdaBoost

Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

```
[[228792      0    2932      0      0      0      0      0      0]
 [   1923      0     167      0      0      0      0      0      0]
 [ 19195      0   27459      0      0      0      0      0      0]
 [    916      0     190      0      0      0      0      0      0]
 [    926      0     222      0      0      0      0      0      0]
 [   1582      0        0      0      0      0      0      0      0]
 [        2      0        0      0      0      0      0      0      0]
 [        4      0        1      0      0      0      0      0      0]
 [   1130      0        2      0      0      0      0      0      0]]
```

Accuracy : 0.8977

Precision : 0.8748

Recall : 0.8977

F1 Score : 0.8797

Classification Report:

	precision	recall	f1-score	support
0	0.90	0.99	0.94	231724
1	0.00	0.00	0.00	2090
2	0.89	0.59	0.71	46654
3	0.00	0.00	0.00	1106
4	0.00	0.00	0.00	1148
5	0.00	0.00	0.00	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.00	0.00	0.00	1132
accuracy			0.90	285443
macro avg	0.20	0.18	0.18	285443

OUTPUT #8: 10. LightGBM

Parameters: test_size = 0.4 | PCA n_components = 10

=====
Confusion Matrix:

```
[[454568    353    2810    1655    3992    182         0     27     79]
 [   508    3545       85         3         5         0         0         0         1]
 [   8079       70   83636     505       75     20         0    520         1]
 [   487     15         2    178    1115         5         0         0    423]
 [   798         2     11     32     813         0         0         0    685]
 [   326         0         2         0     40    2792         0         0         0]
 [         4         0         0         0         0         0         0         0         0]
 [        14         0         0         0         0         0         0         0         0]
 [   1037         0         3         0         0         0         0         0    1383]]
```

Accuracy : 0.9580

Precision : 0.9646

Recall : 0.9580

F1 Score : 0.9609

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.98	0.98	463666
1	0.89	0.85	0.87	4147
2	0.97	0.90	0.93	92906
3	0.08	0.08	0.08	2225
4	0.13	0.35	0.19	2341
5	0.93	0.88	0.91	3160
6	0.00	0.00	0.00	4
7	0.00	0.00	0.00	14
8	0.54	0.57	0.55	2423
accuracy			0.96	570886
macro avg	0.50	0.51	0.50	570886
weighted avg	0.96	0.96	0.96	570886

OUTPUT #9: 11. CatBoost

Parameters: test_size = 0.4 | PCA n_components = 10

=====

Confusion Matrix:

```
[[460978    315    2107    111     65     75      0      0     15]
 [   481   3663         0        3        0        0      0      0      0]
 [   2383        2    90498     12     11        0      0      0      0]
 [   397        1        1   1801     24        0      0      0      1]
 [   584        1        2     21   1733        0      0      0      0]
 [   111        0        2        0        0   3047      0      0      0]
 [     4        0        0        0        0        0      0      0      0]
 [    14        0        0        0        0        0      0      0      0]
 [   1083        0        2        0        0        0      0      0  1338]]
```

Accuracy : 0.9863

Precision : 0.9862

Recall : 0.9863

F1 Score : 0.9859

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	463666
1	0.92	0.88	0.90	4147
2	0.98	0.97	0.98	92906
3	0.92	0.81	0.86	2225
4	0.95	0.74	0.83	2341
5	0.98	0.96	0.97	3160
6	0.00	0.00	0.00	4
7	0.00	0.00	0.00	14
8	0.99	0.55	0.71	2423
accuracy			0.99	570886
macro avg	0.75	0.66	0.69	570886
weighted avg	0.99	0.99	0.99	570886

OUTPUT #7: 9. XGBOOST

Parameters: test_size = 0.4 | PCA n_components = 10

Confusion Matrix:

```
[[461665      91    1645    155     47     15      0      0     48]
 [   239    3902         2      4      0      0      0      0      0]
 [   1573         1   91328         0      4      0      0      0      0]
 [     69         1        0   21130     24      0      0      0      1]
 [    251         1        0     15    2074         0      0      0      0]
 [     29         0        0        0        0   3129      0      0      2]
 [      2         0        0        0        0        0      2      0      0]
 [     10         0        0        0        0        0      0      4      0]
 [    531         0        3        0        0        0      0      0   1889]]
```

Accuracy : 0.9917

Precision : 0.9916

Recall : 0.9917

F1 Score : 0.9916

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	0.99	463666
1	0.98	0.94	0.96	4147
2	0.98	0.98	0.98	92906
3	0.92	0.96	0.94	2225
4	0.97	0.89	0.92	2341
5	1.00	0.99	0.99	3160
6	1.00	0.50	0.67	4
7	1.00	0.29	0.44	14
8	0.97	0.78	0.87	2423
accuracy			0.99	570886
macro avg	0.98	0.81	0.86	570886

OUTPUT #11: 8. Gradient Boosting

Parameters: test_size = 0.6 | PCA n_components = 15

Confusion Matrix:

```
[[688843    358    4495    486    160     80     0    1110    380]
 [   411   5752     11     4     4     0     0     0     0]
 [  5880     4 132427    13    15     7     0    715     1]
 [   150    91    589   2410    94     0     0     0     1]
 [   231    34     3    17   3195     0     0     0     0]
 [   328     0     2     0     0   4408     0     0     3]
 [     4     0     0     0     0     0     0     0     0]
 [    20     0     0     1     1     0     0     0     0]
 [   191     0     6     0     0     1     0     0   3393]]
```

Accuracy : 0.9814

Precision : 0.9834

Recall : 0.9814

F1 Score : 0.9824

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	695912
1	0.92	0.93	0.93	6182
2	0.96	0.95	0.96	139062
3	0.82	0.72	0.77	3335
4	0.92	0.92	0.92	3480
5	0.98	0.93	0.95	4741
6	0.00	0.00	0.00	4
7	0.00	0.00	0.00	22
8	0.90	0.94	0.92	3591
accuracy			0.98	856329
macro avg	0.72	0.71	0.72	856329
weighted avg	0.98	0.98	0.98	856329

OUTPUT #12: 9. XGBoost

Parameters: test_size = 0.6 | PCA n_components = 15

=====

Confusion Matrix:

[[692896	150	2385	223	118	34	0	4	102]
[313	5859	5	5	0	0	0	0	0]
[2126	1	136921	1	13	0	0	0	0]
[143	0	0	3160	31	0	0	0	1]
[95	2	0	19	3364	0	0	0	0]
[35	0	2	0	0	4700	0	0	4]
[2	0	0	0	0	0	2	0	0]
[19	0	0	0	0	0	0	3	0]
[337	0	5	0	0	0	0	0	3249]]

Accuracy : 0.9928

Precision : 0.9928

Recall : 0.9928

F1 Score : 0.9928

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	695912
1	0.97	0.95	0.96	6182
2	0.98	0.98	0.98	139062
3	0.93	0.95	0.94	3335
4	0.95	0.97	0.96	3480
5	0.99	0.99	0.99	4741
6	1.00	0.50	0.67	4
7	0.43	0.14	0.21	22
8	0.97	0.90	0.94	3591
accuracy			0.99	856329
macro avg	0.91	0.82	0.85	856329
weighted avg	0.99	0.99	0.99	856329

OUTPUT #10: 12. AdaBoost

Parameters: test_size = 0.4 | PCA n_components = 10

=====

Confusion Matrix:

[	452997	52	10590	17	10	0	0	0	0]
[	4012	0	36	0	99	0	0	0	0]
[	39062	0	53834	0	10	0	0	0	0]
[	1829	0	396	0	0	0	0	0	0]
[	1872	0	449	20	0	0	0	0	0]
[	3160	0	0	0	0	0	0	0	0]
[	4	0	0	0	0	0	0	0	0]
[	13	0	1	0	0	0	0	0	0]
[	2420	0	3	0	0	0	0	0	0]]

Accuracy : 0.8878

Precision : 0.8622

Recall : 0.8878

F1 Score : 0.8701

Classification Report:

	precision	recall	f1-score	support
0	0.90	0.98	0.93	463666
1	0.00	0.00	0.00	4147
2	0.82	0.58	0.68	92906
3	0.00	0.00	0.00	2225
4	0.00	0.00	0.00	2341
5	0.00	0.00	0.00	3160
6	0.00	0.00	0.00	4
7	0.00	0.00	0.00	14
8	0.00	0.00	0.00	2423
accuracy			0.89	570886
macro avg	0.19	0.17	0.18	570886
weighted avg	0.86	0.89	0.87	570886

OUTPUT #14: 11. CatBoost

Parameters: test_size = 0.6 | PCA n_components = 15

Confusion Matrix:

```
[[690942    514    3784    226    112    139         0         2    193]
 [   681   5478         6     17         0         0         0         0         0]
 [   2974         1 136074         0     13         0         0         0         0]
 [   452         2         0   2836     45         0         0         0         0]
 [   787         1         1     35  2656         0         0         0         0]
 [   354         0         2         0         0   4383         0         0         2]
 [        2         0         0         0         0         0         2         0         0]
 [       22         0         0         0         0         0         0         0         0]
 [   1328         0         4         0         0         0         0         0  2259]]
```

Accuracy : 0.9863

Precision : 0.9862

Recall : 0.9863

F1 Score : 0.9861

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	695912
1	0.91	0.89	0.90	6182
2	0.97	0.98	0.98	139062
3	0.91	0.85	0.88	3335
4	0.94	0.76	0.84	3480
5	0.97	0.92	0.95	4741
6	1.00	0.50	0.67	4
7	0.00	0.00	0.00	22
8	0.92	0.63	0.75	3591
accuracy			0.99	856329
macro avg	0.85	0.72	0.77	856329
weighted avg	0.99	0.99	0.99	856329

OUTPUT #15: 12. AdaBoost

Parameters: test_size = 0.6 | PCA n_components = 15

=====

Confusion Matrix:

[[666893	829	28190	0	0	0	0	0	0]
[5650	0	532	0	0	0	0	0	0]
[22576	374	116112	0	0	0	0	0	0]
[2711	593	31	0	0	0	0	0	0]
[2735	129	616	0	0	0	0	0	0]
[4741	0	0	0	0	0	0	0	0]
[4	0	0	0	0	0	0	0	0]
[19	0	3	0	0	0	0	0	0]
[3587	0	4	0	0	0	0	0	0]]

Accuracy : 0.9144

Precision : 0.8941

Recall : 0.9144

F1 Score : 0.9041

Classification Report:

	precision	recall	f1-score	support
0	0.94	0.96	0.95	695912
1	0.00	0.00	0.00	6182
2	0.80	0.83	0.82	139062
3	0.00	0.00	0.00	3335
4	0.00	0.00	0.00	3480
5	0.00	0.00	0.00	4741
6	0.00	0.00	0.00	4
7	0.00	0.00	0.00	22
8	0.00	0.00	0.00	3591
accuracy			0.91	856329
macro avg	0.19	0.20	0.20	856329
weighted avg	0.89	0.91	0.90	856329

OUTPUT #13: 10. LightGBM

Parameters: test_size = 0.6 | PCA n_components = 15

=====

Confusion Matrix:

[[679479	1253	6750	1455	2219	708	15	434	3599]
[2111	3782	151	42	23	36	0	0	37]
[8396	53	105208	41	148	63	0	6	25147]
[459	6	1691	269	603	102	0	30	175]
[1548	0	50	3	1836	2	0	27	14]
[68	0	2	0	0	4666	0	0	5]
[4	0	0	0	0	0	0	0	0]
[22	0	0	0	0	0	0	0	0]
[672	0	7	0	0	0	0	0	2912]]

Accuracy : 0.9321
Precision : 0.9596
Recall : 0.9321
F1 Score : 0.9431

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.98	0.98	695912
1	0.74	0.61	0.67	6182
2	0.92	0.76	0.83	139062
3	0.15	0.08	0.10	3335
4	0.38	0.53	0.44	3480
5	0.84	0.98	0.90	4741
6	0.00	0.00	0.00	4
7	0.00	0.00	0.00	22
8	0.09	0.81	0.16	3591
accuracy			0.93	856329
macro avg	0.46	0.53	0.46	856329
weighted avg	0.96	0.93	0.94	856329

Accuracy : 0.8631

Precision : 0.8590

Recall : 0.8631

F1 Score : 0.8589

Classification Report:

	precision	recall	f1-score	support
0	0.93	0.90	0.91	31729
1	0.00	0.00	0.00	311
2	0.67	0.85	0.75	7130
3	0.01	0.02	0.01	168
4	0.00	0.00	0.00	178
5	0.00	0.00	0.00	238
6	0.00	0.00	0.00	184
7	0.00	0.00	0.00	44
8	0.00	0.00	0.00	1
9	0.00	0.00	0.00	17
accuracy			0.86	40000
macro avg	0.16	0.18	0.17	40000
weighted avg	0.86	0.86	0.86	40000

Confusion Matrix:

[	31471	15	82	27	4	9	5	57	0	59]
[	72	224	13	0	0	0	0	0	0	2]
[	209	12	6908	0	0	0	0	0	0	1]
[	15	0	0	150	3	0	0	0	0	0]
[	56	0	1	3	118	0	0	0	0	0]
[	8	0	0	0	0	230	0	0	0	0]
[	9	0	0	0	0	0	175	0	0	0]
[	29	0	0	0	0	0	1	1	0	13]
[	1	0	0	0	0	0	0	0	0	0]
[	9	0	1	0	0	0	1	0	0	6]]

Accuracy : 0.9821

Precision : 0.9837

Recall : 0.9821

F1 Score : 0.9826

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	31729
1	0.89	0.72	0.80	311
2	0.99	0.97	0.98	7130
3	0.83	0.89	0.86	168
4	0.94	0.66	0.78	178
5	0.96	0.97	0.96	238
6	0.96	0.95	0.96	184
7	0.02	0.02	0.02	44
8	0.00	0.00	0.00	1
9	0.07	0.35	0.12	17
accuracy			0.98	40000
macro avg	0.67	0.65	0.65	40000
weighted avg	0.98	0.98	0.98	40000

Confusion Matrix:

```
[[31657    12    41     5     1     0     1    12     0     0]
 [   34   273     4     0     0     0     0     0     0     0]
 [   42     9   7079     0     0     0     0     0     0     0]
 [    7     0     0   160     1     0     0     0     0     0]
 [    6     0     0     4   168     0     0     0     0     0]
 [    3     0     0     0     0   235     0     0     0     0]
 [    6     0     0     0     0     0   178     0     0     0]
 [   30     0     0     0     0     0     1    13     0     0]
 [    1     0     0     0     0     0     0     0     0     0]
 [   11     0     0     0     0     0     1     5     0     0]]
```

Accuracy : 0.9941

Precision : 0.9934

Recall : 0.9941

F1 Score : 0.9937

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	31729
1	0.93	0.88	0.90	311
2	0.99	0.99	0.99	7130
3	0.95	0.95	0.95	168
4	0.99	0.94	0.97	178
5	1.00	0.99	0.99	238
6	0.98	0.97	0.98	184
7	0.43	0.30	0.35	44
8	0.00	0.00	0.00	1
9	0.00	0.00	0.00	17
accuracy			0.99	40000
macro avg	0.73	0.70	0.71	40000
weighted avg	0.99	0.99	0.99	40000

Confusion Matrix:

[	[	31638	17	43	10	7	3	4	4	1	2]
[		53	249	9	0	0	0	0	0	0	0]
[		70	0	7058	0	1	1	0	0	0	0]
[		10	0	0	157	1	0	0	0	0	0]
[		12	0	0	2	163	1	0	0	0	0]
[		8	0	0	0	0	230	0	0	0	0]
[		9	0	0	0	0	0	175	0	0	0]
[		18	0	4	0	0	0	1	7	0	14]
[		1	0	0	0	0	0	0	0	0	0]
[		7	0	1	0	0	0	0	6	0	3]]

Accuracy : 0.9920

Precision : 0.9916

Recall : 0.9920

F1 Score : 0.9917

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	1.00	31729
1	0.94	0.80	0.86	311
2	0.99	0.99	0.99	7130
3	0.93	0.93	0.93	168
4	0.95	0.92	0.93	178
5	0.98	0.97	0.97	238
6	0.97	0.95	0.96	184
7	0.41	0.16	0.23	44
8	0.00	0.00	0.00	1
9	0.16	0.18	0.17	17
accuracy			0.99	40000
macro avg	0.73	0.69	0.70	40000
weighted avg	0.99	0.99	0.99	40000

Confusion Matrix:

```
[[30210      79   1440      0      0      0      0      0      0      0]
 [   286      0     25      0      0      0      0      0      0      0]
 [   1012      0   6118      0      0      0      0      0      0      0]
 [    141      0     27      0      0      0      0      0      0      0]
 [    173      0      5      0      0      0      0      0      0      0]
 [    224      0     14      0      0      0      0      0      0      0]
 [    183      0      1      0      0      0      0      0      0      0]
 [     44      0      0      0      0      0      0      0      0      0]
 [      1      0      0      0      0      0      0      0      0      0]
 [     17      0      0      0      0      0      0      0      0      0]]
```

Accuracy : 0.9082

Precision : 0.8850

Recall : 0.9082

F1 Score : 0.8964

Classification Report:

	precision	recall	f1-score	support
0	0.94	0.95	0.94	31729
1	0.00	0.00	0.00	311
2	0.80	0.86	0.83	7130
3	0.00	0.00	0.00	168
4	0.00	0.00	0.00	178
5	0.00	0.00	0.00	238
6	0.00	0.00	0.00	184
7	0.00	0.00	0.00	44
8	0.00	0.00	0.00	1
9	0.00	0.00	0.00	17
accuracy			0.91	40000
macro avg	0.17	0.18	0.18	40000
weighted avg	0.89	0.91	0.90	40000

In [2]:

Confusion Matrix:

```
[[31673      6     26      6      5      2      2      9      0      0]
 [    14    290      7      0      0      0      0      0      0      0]
 [     22      0    7108      0      0      0      0      0      0      0]
 [      3      0      0    163      2      0      0      0      0      0]
 [      1      0      0      4    173      0      0      0      0      0]
 [      3      0      0      0      0    235      0      0      0      0]
 [      9      0      0      0      0      0    174      1      0      0]
 [     27      1      0      0      0      0      1    14      0      1]
 [      1      0      0      0      0      0      0      0      0      0]
 [     11      0      0      0      0      0      1      4      0      1]]
```

Accuracy : 0.9958

Precision : 0.9954

Recall : 0.9958

F1 Score : 0.9955

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	31729
1	0.98	0.93	0.95	311
2	1.00	1.00	1.00	7130
3	0.94	0.97	0.96	168
4	0.96	0.97	0.97	178
5	0.99	0.99	0.99	238
6	0.98	0.95	0.96	184
7	0.50	0.32	0.39	44
8	0.00	0.00	0.00	1
9	0.50	0.06	0.11	17
accuracy			1.00	40000
macro avg	0.78	0.72	0.73	40000
weighted avg	1.00	1.00	1.00	40000

```
[[26050      41    4857     359      39     314      60       5       4       0]
 [      86       1     222       0       1       0       0       1       0       0]
 [   1075     39    5590       0       4      35     43       0     213    131]
 [    139       0      24       0       0       0       1       0       4       0]
 [     72       0     106       0       0       0       0       0       0       0]
 [    207       0      31       0       0       0       0       0       0       0]
 [     83       0     101       0       0       0       0       0       0       0]
 [      6       0      38       0       0       0       0       0       0       0]
 [      1       0       0       0       0       0       0       0       0       0]
 [      1       0      16       0       0       0       0       0       0       0]
```

Accuracy : 0.7910

Precision : 0.8362

Recall : 0.7910

F1 Score : 0.8052

Classification Report:

	precision	recall	f1-score	support
0	0.94	0.82	0.88	31729
1	0.01	0.00	0.01	311
2	0.51	0.78	0.62	7130
3	0.00	0.00	0.00	168
4	0.00	0.00	0.00	178
5	0.00	0.00	0.00	238
6	0.00	0.00	0.00	184
7	0.00	0.00	0.00	44
8	0.00	0.00	0.00	1
9	0.00	0.00	0.00	17
accuracy			0.79	40000
macro avg	0.15	0.16	0.15	40000
weighted avg	0.84	0.79	0.81	40000

In [2]:

Confusion Matrix:

```
[[31670      7     30      5      2      1      1     13      0      0]
 [    19    286      6      0      0      0      0      0      0      0]
 [    30      0    7100      0      0      0      0      0      0      0]
 [      3      0      0    164      1      0      0      0      0      0]
 [      1      0      0      4    173      0      0      0      0      0]
 [      4      0      0      0      0    234      0      0      0      0]
 [      7      0      0      0      0      0    177      0      0      0]
 [     18      1      0      0      0      0      2     21      0      2]
 [      1      0      0      0      0      0      0      0      0      0]
 [      9      0      0      0      0      0      1      6      0      1]]
```

Accuracy : 0.9957

Precision : 0.9953

Recall : 0.9957

F1 Score : 0.9954

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	31729
1	0.97	0.92	0.95	311
2	0.99	1.00	1.00	7130
3	0.95	0.98	0.96	168
4	0.98	0.97	0.98	178
5	1.00	0.98	0.99	238
6	0.98	0.96	0.97	184
7	0.53	0.48	0.50	44
8	0.00	0.00	0.00	1
9	0.33	0.06	0.10	17
accuracy			1.00	40000
macro avg	0.77	0.73	0.74	40000
weighted avg	1.00	1.00	1.00	40000

Confusion Matrix:

[	15739	8	37	7	3	2	3	42	3]
[	26	130	1	0	0	0	0	4	0]
[	64	1	3505	0	0	1	0	2	0]
[	15	0	0	70	1	0	0	0	0]
[	9	0	0	1	79	0	0	0	0]
[	6	0	0	0	0	123	0	0	0]
[	2	0	0	0	0	0	86	0	0]
[	4	0	0	0	0	0	0	12	4]
[	1	0	0	0	0	0	0	7	2]]

Accuracy : 0.9873

Precision : 0.9891

Recall : 0.9873

F1 Score : 0.9880

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	15844
1	0.94	0.81	0.87	161
2	0.99	0.98	0.99	3573
3	0.90	0.81	0.85	86
4	0.95	0.89	0.92	89
5	0.98	0.95	0.96	129
6	0.97	0.98	0.97	88
7	0.18	0.60	0.28	20
8	0.22	0.20	0.21	10
accuracy				0.99
macro avg				0.79
micro avg				0.99
weighted avg				0.99

Confusion Matrix:

[	[	31563	0	165	1	0	0	0	0	0	0]
[		287	18	6	0	0	0	0	0	0	0]
[		1491	0	5639	0	0	0	0	0	0	0]
[		139	0	29	0	0	0	0	0	0	0]
[		142	31	5	0	0	0	0	0	0	0]
[		238	0	0	0	0	0	0	0	0	0]
[		184	0	0	0	0	0	0	0	0	0]
[		44	0	0	0	0	0	0	0	0	0]
[		1	0	0	0	0	0	0	0	0	0]
[		17	0	0	0	0	0	0	0	0	0]]

Accuracy : 0.9305

Precision : 0.9089

Recall : 0.9305

F1 Score : 0.9163

Classification Report:

	precision	recall	f1-score	support
0	0.93	0.99	0.96	31729
1	0.37	0.06	0.10	311
2	0.96	0.79	0.87	7130
3	0.00	0.00	0.00	168
4	0.00	0.00	0.00	178
5	0.00	0.00	0.00	238
6	0.00	0.00	0.00	184
7	0.00	0.00	0.00	44
8	0.00	0.00	0.00	1
9	0.00	0.00	0.00	17
accuracy			0.93	40000
macro avg	0.23	0.18	0.19	40000
weighted avg	0.91	0.93	0.92	40000

Confusion Matrix:

[	15807	5	16	4	0	3	0	7	2]
[	9	149	3	0	0	0	0	0	0]
[	16	0	3557	0	0	0	0	0	0]
[	3	0	0	82	1	0	0	0	0]
[	2	0	0	0	87	0	0	0	0]
[	2	0	0	0	0	127	0	0	0]
[	2	0	0	0	0	0	86	0	0]
[	4	1	0	0	0	0	0	10	5]
[	1	0	0	0	0	0	0	8	1]]

Accuracy : 0.9953

Precision : 0.9954

Recall : 0.9953

F1 Score : 0.9953

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	15844
1	0.96	0.93	0.94	161
2	0.99	1.00	1.00	3573
3	0.95	0.95	0.95	86
4	0.99	0.98	0.98	89
5	0.98	0.98	0.98	129
6	1.00	0.98	0.99	88
7	0.40	0.50	0.44	20
8	0.12	0.10	0.11	10
accuracy			1.00	20000
macro avg	0.82	0.82	0.82	20000
weighted avg	1.00	1.00	1.00	20000

Confusion Matrix:

[[15807	5	16	4	0	3	0	7	2]
[9	149	3	0	0	0	0	0	0]
[16	0	3557	0	0	0	0	0	0]
[3	0	0	82	1	0	0	0	0]
[2	0	0	0	87	0	0	0	0]
[2	0	0	0	0	127	0	0	0]
[2	0	0	0	0	0	86	0	0]
[4	1	0	0	0	0	0	10	5]
[1	0	0	0	0	0	0	8	1]]

Accuracy : 0.9953

Precision : 0.9954

Recall : 0.9953

F1 Score : 0.9953

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	15844
1	0.96	0.93	0.94	161
2	0.99	1.00	1.00	3573
3	0.95	0.95	0.95	86
4	0.99	0.98	0.98	89
5	0.98	0.98	0.98	129
6	1.00	0.98	0.99	88
7	0.40	0.50	0.44	20
8	0.12	0.10	0.11	10
accuracy			1.00	20000
macro avg	0.82	0.82	0.82	20000
weighted avg	1.00	1.00	1.00	20000

Confusion Matrix:

[[15809	6	16	5	0	3	0	5	0]
[9	150	2	0	0	0	0	0	0]
[20	0	3553	0	0	0	0	0	0]
[3	0	0	82	1	0	0	0	0]
[5	0	0	0	84	0	0	0	0]
[2	0	0	0	0	127	0	0	0]
[2	0	0	0	0	0	86	0	0]
[5	1	0	0	0	0	0	12	2]
[1	0	0	0	0	0	0	9	0]]

Accuracy : 0.9951

Precision : 0.9949

Recall : 0.9951

F1 Score : 0.9950

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	15844
1	0.96	0.93	0.94	161
2	0.99	0.99	0.99	3573
3	0.94	0.95	0.95	86
4	0.99	0.94	0.97	89
5	0.98	0.98	0.98	129
6	1.00	0.98	0.99	88
7	0.46	0.60	0.52	20
8	0.00	0.00	0.00	10
accuracy			1.00	20000
macro avg	0.81	0.82	0.82	20000
weighted avg	0.99	1.00	1.00	20000

Confusion Matrix:

[	3150	0	4	0	0	0	0	8	3]
[	3	26	0	0	0	0	0	0	0]
[	11	0	711	0	0	0	0	0	0]
[	1	0	0	15	0	0	0	0	0]
[	0	1	0	0	18	0	0	0	0]
[	0	0	0	0	0	27	0	0	0]
[	1	0	0	0	0	0	16	0	0]
[	3	0	0	0	0	0	0	0	0]
[	2	0	0	0	0	0	0	0	0]]

Accuracy : 0.9908

Precision : 0.9922

Recall : 0.9908

F1 Score : 0.9915

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	0.99	3165
1	0.96	0.90	0.93	29
2	0.99	0.98	0.99	722
3	1.00	0.94	0.97	16
4	1.00	0.95	0.97	19
5	1.00	1.00	1.00	27
6	1.00	0.94	0.97	17
7	0.00	0.00	0.00	3
8	0.00	0.00	0.00	2
accuracy			0.99	4000
macro avg	0.77	0.74	0.76	4000
weighted avg	0.99	0.99	0.99	4000

OUTPUT #1: 8. Gradient Boosting
Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

[	227819	155	1577	548	234	54	303	952	82]
[	176	1765	0	144	2	1	0	0	2]
[	2857	14	43561	27	5	0	0	190	0]
[	158	1	0	933	14	0	0	0	0]
[	189	4	4	42	908	1	0	0	0]
[	153	0	2	0	0	1425	0	0	2]
[	2	0	0	0	0	0	0	0	0]
[	4	0	1	0	0	0	0	0	0]
[	443	0	2	0	0	1	0	0	686]]

Accuracy : 0.9708
Precision : 0.9764
Recall : 0.9708
F1 Score : 0.9733

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.98	0.98	231724
1	0.91	0.84	0.88	2090
2	0.96	0.93	0.95	46654
3	0.55	0.84	0.67	1106
4	0.78	0.79	0.79	1148
5	0.96	0.90	0.93	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.89	0.61	0.72	1132
accuracy			0.97	285443
macro avg	0.67	0.66	0.66	285443
weighted avg	0.98	0.97	0.97	285443

=====

OUTPUT #2: 9. XGBoost

Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

[	229959	89	1449	71	71	57	1	0	27]
[	220	1850	17	2	0	0	0	0	1]
[	1274	0	45372	2	6	0	0	0	0]
[	108	0	1	981	16	0	0	0	0]
[	235	1	2	13	897	0	0	0	0]
[	69	0	0	0	0	1512	0	0	1]
[	1	0	0	0	0	0	1	0	0]
[	3	0	0	0	0	0	0	2	0]
[	445	0	2	0	0	0	0	0	685]]

Accuracy : 0.9853

Precision : 0.9852

Recall : 0.9853

F1 Score : 0.9851

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	231724
1	0.95	0.89	0.92	2090
2	0.97	0.97	0.97	46654
3	0.92	0.89	0.90	1106
4	0.91	0.78	0.84	1148
5	0.96	0.96	0.96	1582
6	0.50	0.50	0.50	2
7	1.00	0.40	0.57	5
8	0.96	0.61	0.74	1132
accuracy			0.99	285443

OUTPUT #1: 8. Gradient Boosting
Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

[	227819	155	1577	548	234	54	303	952	82]
[	176	1765	0	144	2	1	0	0	2]
[	2857	14	43561	27	5	0	0	190	0]
[	158	1	0	933	14	0	0	0	0]
[	189	4	4	42	908	1	0	0	0]
[	153	0	2	0	0	1425	0	0	2]
[	2	0	0	0	0	0	0	0	0]
[	4	0	1	0	0	0	0	0	0]
[	443	0	2	0	0	1	0	0	686]]

Accuracy : 0.9708
Precision : 0.9764
Recall : 0.9708
F1 Score : 0.9733

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.98	0.98	231724
1	0.91	0.84	0.88	2090
2	0.96	0.93	0.95	46654
3	0.55	0.84	0.67	1106
4	0.78	0.79	0.79	1148
5	0.96	0.90	0.93	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.89	0.61	0.72	1132
accuracy			0.97	285443
macro avg	0.67	0.66	0.66	285443
weighted avg	0.98	0.97	0.97	285443

OUTPUT #3: 10. LightGBM

Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

```
[[221702    344    7998    591    548     90      0     21    430]
 [   362   1689      20      0     10      0      0      0      9]
 [   3525     64   42003    953     50      0      0     59      0]
 [    359    136       0    595     14      0      0      2      0]
 [   1103      4     12     10     18      0      0      0      1]
 [    207      0      2      0     10   1363      0      0      0]
 [      1      0      0      1      0      0      0      0      0]
 [      5      0      0      0      0      0      0      0      0]
 [    518     20      3     20    490     33      0      0    48]]
```

Accuracy : 0.9369

Precision : 0.9394

Recall : 0.9369

F1 Score : 0.9377

Classification Report:

	precision	recall	f1-score	support
0	0.97	0.96	0.96	231724
1	0.75	0.81	0.78	2090
2	0.84	0.90	0.87	46654
3	0.27	0.54	0.36	1106
4	0.02	0.02	0.02	1148
5	0.92	0.86	0.89	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.10	0.04	0.06	1132
accuracy			0.94	285443
macro avg	0.43	0.46	0.44	285443

OUTPUT #3: 10. LightGBM

Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

```
[[221702    344    7998    591    548     90      0     21    430]
 [   362   1689      20      0     10      0      0      0      9]
 [   3525     64   42003    953     50      0      0     59      0]
 [   359    136       0    595     14      0      0      2      0]
 [   1103      4     12     10     18      0      0      0      1]
 [   207      0      2      0     10   1363      0      0      0]
 [      1      0      0      1      0      0      0      0      0]
 [      5      0      0      0      0      0      0      0      0]
 [   518     20      3     20    490     33      0      0    48]]
```

Accuracy : 0.9369

Precision : 0.9394

Recall : 0.9369

F1 Score : 0.9377

Classification Report:

	precision	recall	f1-score	support
0	0.97	0.96	0.96	231724
1	0.75	0.81	0.78	2090
2	0.84	0.90	0.87	46654
3	0.27	0.54	0.36	1106
4	0.02	0.02	0.02	1148
5	0.92	0.86	0.89	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.10	0.04	0.06	1132
accuracy			0.94	285443
macro avg	0.43	0.46	0.44	285443

OUTPUT #4: 11. CatBoost

Parameters: test_size = 0.2 | PCA n_components = 5

Confusion Matrix:

```
[[228923    140    2159    108    163    135         0         0     96]
 [   402   1623      54      2      0      8         0         0      1]
 [  2398        0  44224        0     27      1         0         0      4]
 [   317        0        0    773     16        0         0         0      0]
 [   316        1     20      8    803        0         0         0      0]
 [   183        0      3      0      0   1396         0         0      0]
 [      2        0      0      0      0      0         0         0      0]
 [      5        0      0      0      0      0         0         0      0]
 [   494        0      3      0      0      1         0         0   634]]
```

Accuracy : 0.9752

Precision : 0.9748

Recall : 0.9752

F1 Score : 0.9748

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.99	0.99	231724
1	0.92	0.78	0.84	2090
2	0.95	0.95	0.95	46654
3	0.87	0.70	0.77	1106
4	0.80	0.70	0.74	1148
5	0.91	0.88	0.89	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.86	0.56	0.68	1132
accuracy			0.98	285443
macro avg	0.70	0.62	0.65	285443

=====

OUTPUT #2: 9. XGBoost

Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

[	229959	89	1449	71	71	57	1	0	27]
[	220	1850	17	2	0	0	0	0	1]
[	1274	0	45372	2	6	0	0	0	0]
[	108	0	1	981	16	0	0	0	0]
[	235	1	2	13	897	0	0	0	0]
[	69	0	0	0	0	1512	0	0	1]
[	1	0	0	0	0	0	1	0	0]
[	3	0	0	0	0	0	0	2	0]
[	445	0	2	0	0	0	0	0	685]]

Accuracy : 0.9853

Precision : 0.9852

Recall : 0.9853

F1 Score : 0.9851

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	231724
1	0.95	0.89	0.92	2090
2	0.97	0.97	0.97	46654
3	0.92	0.89	0.90	1106
4	0.91	0.78	0.84	1148
5	0.96	0.96	0.96	1582
6	0.50	0.50	0.50	2
7	1.00	0.40	0.57	5
8	0.96	0.61	0.74	1132
accuracy			0.99	285443

=====

OUTPUT #5: 12. AdaBoost

Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

```
[[228792      0    2932      0      0      0      0      0      0]
 [   1923      0     167      0      0      0      0      0      0]
 [ 19195      0   27459      0      0      0      0      0      0]
 [    916      0     190      0      0      0      0      0      0]
 [    926      0     222      0      0      0      0      0      0]
 [   1582      0        0      0      0      0      0      0      0]
 [        2      0        0      0      0      0      0      0      0]
 [        4      0        1      0      0      0      0      0      0]
 [   1130      0        2      0      0      0      0      0      0]]
```

Accuracy : 0.8977

Precision : 0.8748

Recall : 0.8977

F1 Score : 0.8797

Classification Report:

	precision	recall	f1-score	support
0	0.90	0.99	0.94	231724
1	0.00	0.00	0.00	2090
2	0.89	0.59	0.71	46654
3	0.00	0.00	0.00	1106
4	0.00	0.00	0.00	1148
5	0.00	0.00	0.00	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.00	0.00	0.00	1132
accuracy			0.90	285443
macro avg	0.20	0.18	0.18	285443

=====

OUTPUT #5: 12. AdaBoost

Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

```
[[228792      0    2932      0      0      0      0      0      0]
 [   1923      0     167      0      0      0      0      0      0]
 [ 19195      0   27459      0      0      0      0      0      0]
 [    916      0     190      0      0      0      0      0      0]
 [    926      0     222      0      0      0      0      0      0]
 [   1582      0        0      0      0      0      0      0      0]
 [        2      0        0      0      0      0      0      0      0]
 [        4      0        1      0      0      0      0      0      0]
 [   1130      0        2      0      0      0      0      0      0]]
```

Accuracy : 0.8977

Precision : 0.8748

Recall : 0.8977

F1 Score : 0.8797

Classification Report:

	precision	recall	f1-score	support
0	0.90	0.99	0.94	231724
1	0.00	0.00	0.00	2090
2	0.89	0.59	0.71	46654
3	0.00	0.00	0.00	1106
4	0.00	0.00	0.00	1148
5	0.00	0.00	0.00	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.00	0.00	0.00	1132
accuracy			0.90	285443
macro avg	0.20	0.18	0.18	285443

OUTPUT #4: 11. CatBoost

Parameters: test_size = 0.2 | PCA n_components = 5

Confusion Matrix:

```
[[228923    140    2159    108    163    135         0         0     96]
 [   402   1623        54         2         0         8         0         0         1]
 [  2398         0  44224         0        27         1         0         0         4]
 [   317         0         0    773        16         0         0         0         0]
 [   316         1        20         8    803         0         0         0         0]
 [   183         0         3         0         0   1396         0         0         0]
 [     2         0         0         0         0         0         0         0         0]
 [     5         0         0         0         0         0         0         0         0]
 [   494         0         3         0         0         1         0         0    634]]
```

Accuracy : 0.9752

Precision : 0.9748

Recall : 0.9752

F1 Score : 0.9748

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.99	0.99	231724
1	0.92	0.78	0.84	2090
2	0.95	0.95	0.95	46654
3	0.87	0.70	0.77	1106
4	0.80	0.70	0.74	1148
5	0.91	0.88	0.89	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.86	0.56	0.68	1132
accuracy			0.98	285443
macro avg	0.70	0.62	0.65	285443

OUTPUT #7: 9. XGBOOST

Parameters: test_size = 0.4 | PCA n_components = 10

Confusion Matrix:

```
[[461665      91    1645    155     47     15      0      0     48]
 [   239    3902         2      4      0      0      0      0      0]
 [   1573         1   91328         0      4      0      0      0      0]
 [    69         1      0   21130     24      0      0      0      1]
 [   251         1      0     15    2074         0      0      0      0]
 [    29         0      0      0      0   3129      0      0      2]
 [     2         0      0      0      0      0      2      0      0]
 [    10         0      0      0      0      0      0      4      0]
 [   531         0      3      0      0      0      0      0   1889]]
```

Accuracy : 0.9917

Precision : 0.9916

Recall : 0.9917

F1 Score : 0.9916

Classification Report:

	precision	recall	f1-score	support
0	0.99	1.00	0.99	463666
1	0.98	0.94	0.96	4147
2	0.98	0.98	0.98	92906
3	0.92	0.96	0.94	2225
4	0.97	0.89	0.92	2341
5	1.00	0.99	0.99	3160
6	1.00	0.50	0.67	4
7	1.00	0.29	0.44	14
8	0.97	0.78	0.87	2423
accuracy			0.99	570886
macro avg	0.98	0.81	0.86	570886

OUTPUT #8: 10. LightGBM

Parameters: test_size = 0.4 | PCA n_components = 10

=====
Confusion Matrix:

```
[[454568    353    2810    1655    3992    182         0     27     79]
 [   508    3545         85         3         5         0         0         0         1]
 [   8079         70   83636     505         75         20         0    520         1]
 [   487         15         2     178    1115         5         0         0    423]
 [   798         2        11        32     813         0         0         0    685]
 [   326         0         2         0        40    2792         0         0         0]
 [         4         0         0         0         0         0         0         0         0]
 [        14         0         0         0         0         0         0         0         0]
 [   1037         0         3         0         0         0         0         0    1383]]
```

Accuracy : 0.9580

Precision : 0.9646

Recall : 0.9580

F1 Score : 0.9609

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.98	0.98	463666
1	0.89	0.85	0.87	4147
2	0.97	0.90	0.93	92906
3	0.08	0.08	0.08	2225
4	0.13	0.35	0.19	2341
5	0.93	0.88	0.91	3160
6	0.00	0.00	0.00	4
7	0.00	0.00	0.00	14
8	0.54	0.57	0.55	2423
accuracy			0.96	570886
macro avg	0.50	0.51	0.50	570886
weighted avg	0.96	0.96	0.96	570886

OUTPUT #6: 8. Gradient Boosting

Parameters: test_size = 0.4 | PCA n_components = 10

Confusion Matrix:

```
[[458655    275    3493    293    104    103         0    571    172]
 [   180   3958         6         3         0         0         0         0         0]
 [   4162         3   88643         8         4         0         0        86         0]
 [   156         3        12   1982        63         0         0         0         9]
 [   449         1         1        13   1876         1         0         0         0]
 [    84         0         2         0         0   3072         0         0         2]
 [     4         0         0         0         0         0         0         0         0]
 [    14         0         0         0         0         0         0         0         0]
 [   399         0         2         0         0         1         0         0    2021]]
```

Accuracy : 0.9813

Precision : 0.9823

Recall : 0.9813

F1 Score : 0.9818

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	463666
1	0.93	0.95	0.94	4147
2	0.96	0.95	0.96	92906
3	0.86	0.89	0.88	2225
4	0.92	0.80	0.86	2341
5	0.97	0.97	0.97	3160
6	0.00	0.00	0.00	4
7	0.00	0.00	0.00	14
8	0.92	0.83	0.87	2423
accuracy			0.98	570886
macro avg	0.73	0.71	0.72	570886

OUTPUT #10: 12. AdaBoost

Parameters: test_size = 0.4 | PCA n_components = 10

=====

Confusion Matrix:

[	452997	52	10590	17	10	0	0	0	0]
[	4012	0	36	0	99	0	0	0	0]
[	39062	0	53834	0	10	0	0	0	0]
[	1829	0	396	0	0	0	0	0	0]
[	1872	0	449	20	0	0	0	0	0]
[	3160	0	0	0	0	0	0	0	0]
[	4	0	0	0	0	0	0	0	0]
[	13	0	1	0	0	0	0	0	0]
[	2420	0	3	0	0	0	0	0	0]]

Accuracy : 0.8878

Precision : 0.8622

Recall : 0.8878

F1 Score : 0.8701

Classification Report:

	precision	recall	f1-score	support
0	0.90	0.98	0.93	463666
1	0.00	0.00	0.00	4147
2	0.82	0.58	0.68	92906
3	0.00	0.00	0.00	2225
4	0.00	0.00	0.00	2341
5	0.00	0.00	0.00	3160
6	0.00	0.00	0.00	4
7	0.00	0.00	0.00	14
8	0.00	0.00	0.00	2423
accuracy			0.89	570886
macro avg	0.19	0.17	0.18	570886
weighted avg	0.86	0.89	0.87	570886

OUTPUT #11: 8. Gradient Boosting

Parameters: test_size = 0.6 | PCA n_components = 15

Confusion Matrix:

```
[[688843    358    4495    486    160     80     0    1110    380]
 [   411   5752     11     4     4     0     0     0     0]
 [  5880     4 132427    13    15     7     0    715     1]
 [   150    91    589   2410    94     0     0     0     1]
 [   231    34     3    17   3195     0     0     0     0]
 [   328     0     2     0     0   4408     0     0     3]
 [     4     0     0     0     0     0     0     0     0]
 [    20     0     0     1     1     0     0     0     0]
 [   191     0     6     0     0     1     0     0   3393]]
```

Accuracy : 0.9814

Precision : 0.9834

Recall : 0.9814

F1 Score : 0.9824

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	695912
1	0.92	0.93	0.93	6182
2	0.96	0.95	0.96	139062
3	0.82	0.72	0.77	3335
4	0.92	0.92	0.92	3480
5	0.98	0.93	0.95	4741
6	0.00	0.00	0.00	4
7	0.00	0.00	0.00	22
8	0.90	0.94	0.92	3591
accuracy			0.98	856329
macro avg	0.72	0.71	0.72	856329
weighted avg	0.98	0.98	0.98	856329

OUTPUT #9: 11. CatBoost

Parameters: test_size = 0.4 | PCA n_components = 10

=====

Confusion Matrix:

```
[[460978    315    2107    111     65     75      0      0     15]
 [   481   3663         0        3        0        0      0      0      0]
 [   2383        2    90498     12     11        0      0      0      0]
 [   397        1        1   1801     24        0      0      0      1]
 [   584        1        2     21   1733        0      0      0      0]
 [   111        0        2        0        0   3047      0      0      0]
 [     4        0        0        0        0        0      0      0      0]
 [    14        0        0        0        0        0      0      0      0]
 [   1083        0        2        0        0        0      0      0  1338]]
```

Accuracy : 0.9863

Precision : 0.9862

Recall : 0.9863

F1 Score : 0.9859

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	463666
1	0.92	0.88	0.90	4147
2	0.98	0.97	0.98	92906
3	0.92	0.81	0.86	2225
4	0.95	0.74	0.83	2341
5	0.98	0.96	0.97	3160
6	0.00	0.00	0.00	4
7	0.00	0.00	0.00	14
8	0.99	0.55	0.71	2423
accuracy			0.99	570886
macro avg	0.75	0.66	0.69	570886
weighted avg	0.99	0.99	0.99	570886

OUTPUT #13: 10. LightGBM

Parameters: test_size = 0.6 | PCA n_components = 15

=====

Confusion Matrix:

[[679479	1253	6750	1455	2219	708	15	434	3599]
[2111	3782	151	42	23	36	0	0	37]
[8396	53	105208	41	148	63	0	6	25147]
[459	6	1691	269	603	102	0	30	175]
[1548	0	50	3	1836	2	0	27	14]
[68	0	2	0	0	4666	0	0	5]
[4	0	0	0	0	0	0	0	0]
[22	0	0	0	0	0	0	0	0]
[672	0	7	0	0	0	0	0	2912]]

Accuracy : 0.9321
Precision : 0.9596
Recall : 0.9321
F1 Score : 0.9431

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.98	0.98	695912
1	0.74	0.61	0.67	6182
2	0.92	0.76	0.83	139062
3	0.15	0.08	0.10	3335
4	0.38	0.53	0.44	3480
5	0.84	0.98	0.90	4741
6	0.00	0.00	0.00	4
7	0.00	0.00	0.00	22
8	0.09	0.81	0.16	3591
accuracy			0.93	856329
macro avg	0.46	0.53	0.46	856329
weighted avg	0.96	0.93	0.94	856329

OUTPUT #14: 11. CatBoost

Parameters: test_size = 0.6 | PCA n_components = 15

Confusion Matrix:

```
[[690942    514    3784    226    112    139         0         2    193]
 [   681   5478         6     17         0         0         0         0         0]
 [   2974         1 136074         0     13         0         0         0         0]
 [   452         2         0   2836     45         0         0         0         0]
 [   787         1         1     35  2656         0         0         0         0]
 [   354         0         2         0         0   4383         0         0         2]
 [         2         0         0         0         0         0         2         0         0]
 [        22         0         0         0         0         0         0         0         0]
 [   1328         0         4         0         0         0         0         0  2259]]
```

Accuracy : 0.9863

Precision : 0.9862

Recall : 0.9863

F1 Score : 0.9861

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	695912
1	0.91	0.89	0.90	6182
2	0.97	0.98	0.98	139062
3	0.91	0.85	0.88	3335
4	0.94	0.76	0.84	3480
5	0.97	0.92	0.95	4741
6	1.00	0.50	0.67	4
7	0.00	0.00	0.00	22
8	0.92	0.63	0.75	3591
accuracy			0.99	856329
macro avg	0.85	0.72	0.77	856329
weighted avg	0.99	0.99	0.99	856329

OUTPUT #15: 12. AdaBoost

Parameters: test_size = 0.6 | PCA n_components = 15

=====

Confusion Matrix:

[[666893	829	28190	0	0	0	0	0	0]
[5650	0	532	0	0	0	0	0	0]
[22576	374	116112	0	0	0	0	0	0]
[2711	593	31	0	0	0	0	0	0]
[2735	129	616	0	0	0	0	0	0]
[4741	0	0	0	0	0	0	0	0]
[4	0	0	0	0	0	0	0	0]
[19	0	3	0	0	0	0	0	0]
[3587	0	4	0	0	0	0	0	0]]

Accuracy : 0.9144

Precision : 0.8941

Recall : 0.9144

F1 Score : 0.9041

Classification Report:

	precision	recall	f1-score	support
0	0.94	0.96	0.95	695912
1	0.00	0.00	0.00	6182
2	0.80	0.83	0.82	139062
3	0.00	0.00	0.00	3335
4	0.00	0.00	0.00	3480
5	0.00	0.00	0.00	4741
6	0.00	0.00	0.00	4
7	0.00	0.00	0.00	22
8	0.00	0.00	0.00	3591
accuracy			0.91	856329
macro avg	0.19	0.20	0.20	856329
weighted avg	0.89	0.91	0.90	856329

OUTPUT #12: 9. XGBoost

Parameters: test_size = 0.6 | PCA n_components = 15

=====

Confusion Matrix:

[[692896	150	2385	223	118	34	0	4	102]
[313	5859	5	5	0	0	0	0	0]
[2126	1	136921	1	13	0	0	0	0]
[143	0	0	3160	31	0	0	0	1]
[95	2	0	19	3364	0	0	0	0]
[35	0	2	0	0	4700	0	0	4]
[2	0	0	0	0	0	2	0	0]
[19	0	0	0	0	0	0	3	0]
[337	0	5	0	0	0	0	0	3249]]

Accuracy : 0.9928
Precision : 0.9928
Recall : 0.9928
F1 Score : 0.9928

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	695912
1	0.97	0.95	0.96	6182
2	0.98	0.98	0.98	139062
3	0.93	0.95	0.94	3335
4	0.95	0.97	0.96	3480
5	0.99	0.99	0.99	4741
6	1.00	0.50	0.67	4
7	0.43	0.14	0.21	22
8	0.97	0.90	0.94	3591
accuracy			0.99	856329
macro avg	0.91	0.82	0.85	856329
weighted avg	0.99	0.99	0.99	856329

OUTPUT #1: 8. Gradient Boosting
Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

[	227819	155	1577	548	234	54	303	952	82]
[	176	1765	0	144	2	1	0	0	2]
[	2857	14	43561	27	5	0	0	190	0]
[	158	1	0	933	14	0	0	0	0]
[	189	4	4	42	908	1	0	0	0]
[	153	0	2	0	0	1425	0	0	2]
[	2	0	0	0	0	0	0	0	0]
[	4	0	1	0	0	0	0	0	0]
[	443	0	2	0	0	1	0	0	686]]

Accuracy : 0.9708
Precision : 0.9764
Recall : 0.9708
F1 Score : 0.9733

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.98	0.98	231724
1	0.91	0.84	0.88	2090
2	0.96	0.93	0.95	46654
3	0.55	0.84	0.67	1106
4	0.78	0.79	0.79	1148
5	0.96	0.90	0.93	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.89	0.61	0.72	1132
accuracy			0.97	285443
macro avg	0.67	0.66	0.66	285443
weighted avg	0.98	0.97	0.97	285443

OUTPUT #1: 8. Gradient Boosting
Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

[	227819	155	1577	548	234	54	303	952	82]
[	176	1765	0	144	2	1	0	0	2]
[	2857	14	43561	27	5	0	0	190	0]
[	158	1	0	933	14	0	0	0	0]
[	189	4	4	42	908	1	0	0	0]
[	153	0	2	0	0	1425	0	0	2]
[	2	0	0	0	0	0	0	0	0]
[	4	0	1	0	0	0	0	0	0]
[	443	0	2	0	0	1	0	0	686]]

Accuracy : 0.9708
Precision : 0.9764
Recall : 0.9708
F1 Score : 0.9733

Classification Report:

	precision	recall	f1-score	support
0	0.98	0.98	0.98	231724
1	0.91	0.84	0.88	2090
2	0.96	0.93	0.95	46654
3	0.55	0.84	0.67	1106
4	0.78	0.79	0.79	1148
5	0.96	0.90	0.93	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.89	0.61	0.72	1132
accuracy			0.97	285443
macro avg	0.67	0.66	0.66	285443
weighted avg	0.98	0.97	0.97	285443

=====

OUTPUT #2: 9. XGBoost

Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

[	229959	89	1449	71	71	57	1	0	27]
[	220	1850	17	2	0	0	0	0	1]
[	1274	0	45372	2	6	0	0	0	0]
[	108	0	1	981	16	0	0	0	0]
[	235	1	2	13	897	0	0	0	0]
[	69	0	0	0	0	1512	0	0	1]
[	1	0	0	0	0	0	1	0	0]
[	3	0	0	0	0	0	0	2	0]
[	445	0	2	0	0	0	0	0	685]]

Accuracy : 0.9853

Precision : 0.9852

Recall : 0.9853

F1 Score : 0.9851

Classification Report:

	precision	recall	f1-score	support
0	0.99	0.99	0.99	231724
1	0.95	0.89	0.92	2090
2	0.97	0.97	0.97	46654
3	0.92	0.89	0.90	1106
4	0.91	0.78	0.84	1148
5	0.96	0.96	0.96	1582
6	0.50	0.50	0.50	2
7	1.00	0.40	0.57	5
8	0.96	0.61	0.74	1132
accuracy			0.99	285443

OUTPUT #3: 10. LightGBM

Parameters: test_size = 0.2 | PCA n_components = 5

=====

Confusion Matrix:

```
[[221702    344    7998    591    548     90      0     21    430]
 [   362   1689      20      0     10      0      0      0      9]
 [   3525     64   42003    953     50      0      0     59      0]
 [    359    136       0    595     14      0      0      2      0]
 [   1103      4     12     10     18      0      0      0      1]
 [    207      0      2      0     10   1363      0      0      0]
 [      1      0      0      1      0      0      0      0      0]
 [      5      0      0      0      0      0      0      0      0]
 [    518     20      3     20    490     33      0      0    48]]
```

Accuracy : 0.9369

Precision : 0.9394

Recall : 0.9369

F1 Score : 0.9377

Classification Report:

	precision	recall	f1-score	support
0	0.97	0.96	0.96	231724
1	0.75	0.81	0.78	2090
2	0.84	0.90	0.87	46654
3	0.27	0.54	0.36	1106
4	0.02	0.02	0.02	1148
5	0.92	0.86	0.89	1582
6	0.00	0.00	0.00	2
7	0.00	0.00	0.00	5
8	0.10	0.04	0.06	1132
accuracy			0.94	285443
macro avg	0.43	0.46	0.44	285443